

MATH0041 Mathematics for Science 2

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0041
<i>Level:</i>	4 (UG)
<i>Normal student group(s):</i>	UG: Y1 Mathematics programmes
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	2
<i>Assessment:</i>	85% examination, 15% coursework
<i>Normal Pre-requisites:</i>	MATH0040 (Mathematics for Science 1) NSCI0005
<i>Lecturer:</i>	Prof E Burman

Course Description and Objectives

The traditional title for this material is Advanced Calculus and Geometry. Building on the material covered in MATH0040, it provides a foundation course in 3-dimensional geometry, calculus of several variables, differential operators and eigenvalue problems.

Recommended Texts

There are many excellent textbooks covering this material. One particularly suitable one is the following:

- ML Boas, *Mathematical Methods in the Physical Sciences* (Wiley)

Detailed Syllabus

- Introduction to matrices, matrix multiplication and addition, inverses and determinants.
- Functions of several variables. Change of coordinates and Chain Rule. Critical points of functions of 2 variables; maxima, minima, saddle points.
- Linear differential operators and Heisenberg (= commutator) bracket.
- Vector fields. Normal and tangent fields to a surface. Div, grad, curl. Laplacian in spherical and cylindrical coordinates.
- Row-Echelon Form, solving systems of linear equations, eigenvectors and eigenvalues. Diagonalisation of 3×3 symmetric matrices.
- Orthonormal sets of vectors. Orthonormal sets of functions. Fourier series.